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Agentic AI Approach to Data Dimension Reduction

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Abstract

Three Statistical Data Dimension Reduction techniques are applied to macroeconomics and financial time series. These are Spearman Correlation, Cluster Analysis and Principal Component Analysis (PCA). We developed an AI Agent solution, with underlying codes written in Python, version 3.14, that utilizes those three methodologies widely adopted in the financial industry and other areas, like health care, media, supply chain, etc.

A Statistical significance criterion, mimicking the Student *t test*, is introduced as a basis for the Agent's decision-making for the estimated Spearman correlations. A *Judge* AI-Agent combines statistical analytics with specific business requirements. Overall, we suggest that our AI Agent solution could be a tremendous boost for practitioners and researchers who routinely analyze massive data sets for defined purposes.

Key Words: Dimensionality Reduction, Correlations, Cluster, PCA, AI-Agents

Introduction

There is a need to identify dominant macroeconomic and financial drivers to predict businesses' revenue or growth and other key metrics in the marketplace. Most financial institutions routinely perform this exercise for regulatory Stress Test, Sensitivity Analyses, projection of short term, medium- and long-term business strategies and operations, in daily, quarterly and yearly reports. Their approaches span rudimentary analytics and basic extrapolations, to sophisticated Statistical prediction and forecasting models.

In the universe of Big Data at hand today, there is a **primary challenge** to be able to identify a meaningful subset of data for subsequent analytics. In effect, this is a Data Dimension Reduction exercise that requires robust Statistical techniques. There are many competing techniques like Cluster Analysis (K-Means, 2-Step, Kohonen) to find patterns; Correlations, to estimate various associations; Factor Analysis

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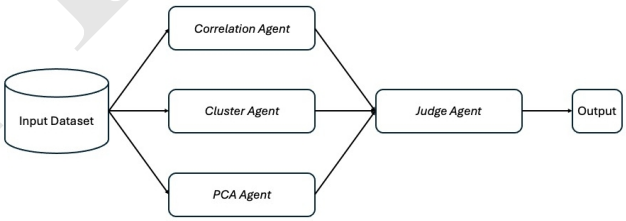
for dimension reduction and assuming that the covariation in the observed variables is due to the presence of one or more latent variables; Decision Trees (CHAID, C5.0, C&RT) to find patterns related to known outcome; Logistic Regression to find patterns related to a known categorical outcome; Linear Regression to find patterns related to a known continuous numerical outcome; Market Basket Analysis to utilize some known selection criteria; Text Analysis to find main ideas in blocks of texts such as program or network descriptions; Correspondence Analysis (CA), a chi-square based descriptive technique that traditionally applied to contingency tables, and Principal Component Analysis (PCA) for data dimension reduction. In the present work, Spearman Correlation, Cluster Analysis, and PCA are utilized because of their widespread use in the marketplace.

Financial institutions and other market actors are obligated to frequently report defined key metrics. Due to the repetitiveness of the data dimension reduction works, **a second challenge** appears to be how to accurately and efficiently automate all involved processes and perspectives. Oke and Lyhagen (1999) suggested some automated processes for Stationarity tests in macroeconomics time series data. Okpala, Golgoon and Ravi Kannan (2025) proposed Agentic AI systems for financial models and Model Risk Management (MRM). Ding H. *et al* (2024) published a systematic review of relevant works that leverage Large Language Models (LLM) trading agents. In this research, we are presenting an AI Agent framework that will not only conduct automated statistical analyses, but also integrates knowledge tailored to the business units such as risk management, compliance, finance, treasury, trading floor, commercial and retail banking; see for example Oke, Grossman *et al* (2026) on AI Adoption in critical areas of Market Risk. Through prompt engineering we have also designed a *Judge* Agent that can interpret statistical outputs in light of the business unit's objectives, risk appetite, performance measures, and reporting requirements.

Methodology

The methodology employed includes five components: Input Data, Spearman Correlation, Cluster Analysis, PCA, and a *Judge* Agent, with respective illustrations and descriptions. For a clear grasp of the work process, the Agents Workflow is shown in **Figure 1** below.

Figure 1: Dimension Reduction AI Agents Workflow



The user provides a dataset, identifies the dependent variable within the dataset, and supplies any additional business context, objectives, and or goals. Three statistical analyses are performed in parallel. Next, their outputs are evaluated by an independent judge agent which considers both the statistical results and the business context to generate a final recommendation.

The agentic workflow begins with a dataset input. Each agent performs a multitude of steps prior to conducting its statistical analysis to ensure the dataset is adequately prepared. This included identifying

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all variable types, retaining numeric non-index type columns, and removing rows with missing values. The three analysis agents operate in parallel as each statistical approach is independent of the other. The output provided to the user, includes the pure analyses results and a recommendation for the final retained variable set.

Data

The basis for this research is a macro-financial dataset comprising $n = 26$ annual observations on 10 macroeconomic and 1 financial variables, hence $p = 11$, sourced from the Federal Reserve Economic Data (FRED) database, created and maintained by the Research Department at the Federal Reserve Bank of St. Louis. The 10 macroeconomic variables, the independent variable, span four economic domains; Real Economic Activities, Prices, Interest Rates, Assets. Updated sources of the data are listed in References. The Financial variable (Firms' Revenue), the dependent variable, is sourced from S&P Capital IQ and is tailored to mimic the Americas division credit portfolio of an European Investment Bank. **Table 1** exhibits details on the variables, such as Domain, Descriptive Names, and Transformations or Units.

Table 1: Macroeconomic and Financial Time Series

Domain	Variable	Transformation / (Unit)
Real activity	Real gross domestic product (GDP)	Year-over-year growth rate (%)
Real activity	Unemployment rate	Year-over-year change (percentage points)
Real activity	S&P Cotality Case-Shiller U.S. National Home Price Index	Year-over-year growth rate (%)
Pricing	Sticky-price consumer price index (CPI), excluding food and energy	Year-over-year growth rate (%)
Pricing	West Texas Intermediate (WTI) crude oil price	Year-over-year growth rate (%)
Interest rates and credit	3-month Treasury bill rate	Year-over-year change (percentage points)
Interest rates and credit	5-year Treasury yield	Year-over-year change (percentage points)
Interest rates and credit	10-year Treasury yield	Year-over-year change (percentage points)
Interest rates and credit	ICE BofA BBB U.S. corporate index effective yield	Year-over-year change (percentage points)
Assets markets	S&P 500 index	Year-over-year growth rate (%)
Firm	Median annual revenue across sampled	Year-over-year growth rate (%)

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performance	firms	

Note: ICE BofA is the Intercontinental Exchange Bank of America; BBB is an investment-grade corporate-credit rating category; S&P is Standard & Poor's.
Source. Authors' calculations using data from FRED and S&P Capital IQ Pro.

For all the variables in **Table 1** above the Yearly Growth Rate series are created as follows:

$$\text{Macroeconomic Indicator (MI) Growth Rate}_{Year} = \frac{MI_{Year} - \text{MI}_{Year-1}}{|\text{MI}_{Year-1}|} \quad (1)$$

Where the change from the previous year is:

$$\text{Macroeconomic Indicator Change}_{Year} = \text{MI}_{Year} - \text{MI}_{Year-1} \quad (2)$$

Financial Variables (FV) are also transformed into annual growth rates and an aggregation is performed to construct a single proxy representative of the portfolio segment. The median is used as the aggregation statistic because unlike the mean, the median is less sensitive to a small number of firms experiencing exceptionally high or low growth, which may reflect idiosyncratic factors rather than systematic macroeconomic effects; see Huber & Ronchetti (2009). This approach implicitly assumes that firms within a segment share a common macroeconomic component in their growth dynamics, even if the magnitude of the impact varies at the firm level. FV is obtained as follows:

$$\text{Financial Variable (FV) Growth Rate}_{Year} = \frac{FV_{Year} - \text{FV}_{Year-1}}{|\text{FV}_{Year-1}|} \quad (3)$$

Spearman Correlation

After data preparation, the *Correlation* Agent computes a Spearman rank-order correlation, Spearman (1904), matrix across the cleaned dataset, including the dependent and all independent variables. Unlike the Pearson correlation that produces the strength and direction of the linear relationship between two variables, the Spearman method measures the strength and direction of the monotonic relationship between every pair of analytical variables. Generally, the correlation ρ is calculated through the following Pearson formula, irrespective of the scale of the data; continuous, or ordinal and ranked:

$$\rho = \frac{\sum_{i=1}^{n}(X_i - \underline{X})(Y_i - \underline{Y})}{\sqrt{\sum_{i=1}^{n}(X_i - \underline{X})^2 \sum_{i=1}^{n}(Y_i - \underline{Y})^2}} \quad (4)$$

Where X_i and Y_i are the two variables being compared; $\underline{X}$ and $\underline{Y}$ represent their mean values respectively.

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Given that the time-series dataset contains 26 complete observations, a Student's *t*-distribution approximation, see Evans, Hasting and Peacock (2000) is used to establish the minimum correlation associated with statistical significance; Zar (1972).

$$t = \frac{\rho_s \sqrt{n-2}}{\sqrt{1-\rho_s^2}} \quad (5)$$

Where n is the number of complete observations and $df = n - 2$.

The two-tailed critical t value at $\alpha = 0.05$ is then converted to a corresponding critical correlation. An independent variable is retained when the absolute value of its Spearman correlation is greater than or equal to the critical correlation, which is used as an approximation for statistical significance.

Hence, as illustrated in Spearman's Rank Correlation Critical Values Table, Two-Tailed: R_s critical (n=26, Two-Tailed $\alpha = 0.05$) = 0.390. Therefore, at n = 26, two-tailed $\alpha = 0.05$: reject H_0 if $|\rho_s| \geq 0.390$. Fail to reject H_0 if $|\rho_s| < 0.390$; where H_0 would suggest that the observed correlation occurred by chance.

Cluster Analysis

Cluster Analysis is a powerful statistical technique to group similar data points, according to specific criteria. For a comprehensive theoretical overview, see Mardia, Kent and Bibby (1989) p. 360-386. Here, it is applied to identify the most representative macroeconomic variables, as candidates for insights in firms' revenue.

To approach cluster analysis, the independent variables are separated from the dependent variable. The absolute values of the Spearman rank correlations are used to calculate the distance matrix. Agglomerative hierarchical clustering was performed on the precomputed predictor distance matrix using complete linkage in *scikit-learn Python* package; see Virtanen et al., 2020. Hence, for clusters *A* and *B*, the distance $D(A,B)$ is:

$$D(A,B) = \max\{d(x,y) : x \in A, y \in B\} \quad (6)$$

Where A and B are two separate clusters, x is a data point in Cluster A, y is a data point in Cluster B, and $d(x,y)$ represents the distance as defined by the precomputed predictor distance matrix.

The agent selects the optimal number of clusters, *k*, by calculating the mean silhouette score across candidate solutions as:

$$S(k) = \frac{1}{n}\sum_{i=1}^{n} s(i) \quad (7)$$

And

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$$s(\mathrm{i}) = \frac{b(\mathrm{i}) - a(\mathrm{i})}{max(a(\mathrm{i}),b(\mathrm{i}))} \quad (8)$$

Where *n* is the number of observations, $a(\mathrm{i})$ is the average distance from feature i to all other features in its assigned cluster, and $b(\mathrm{i})$ is the average distance from feature i to all other features in its closest neighboring cluster. The value of k with the highest silhouette score is retained, as suggested by Rousseeuw (1987).

Each cluster's absolute minimum within-cluster Spearman correlation, medoid, and maximum-target-correlation representative variable sets are defined. The medoid is the variable in the cluster that is most centrally representative of the other predictors in its cluster, whereas the maximum is the variable with the highest absolute Spearman correlation with the dependent variable. For each reduced feature set, the Variance Inflation Factor is calculated using an intercept for an additional diagnostic of remaining multicollinearity. Hence, the choice of a feature set relies on the combination of these multiple statistical criteria.

Principal Component Analysis

PCA is a robust statistical technique to analyze data, identify patterns in the data, then compress the data, i.e., by reducing the number of dimensions, without much loss of information. This technique is widely used in image compression. As the first objective, PCA seeks in the data the Standardized Linear Combination (SLC) of the original variables which have maximal variance. More generally, PCA looks for a few linear combinations which can be used to summarize the data, losing in the process as little information as possible; see Mardia, Kent, and Bibby (1989), p 213 - 246.

When variables differ materially in magnitude and are measured on different scales, like the macroeconomic dataset in question, it is common to standardize the variables. This ensures that all input variables contribute proportionately to the analysis regardless of their original scale or units of measurements. In this instance, standardized macroeconomic variables create a data matrix: $Z \in R^{n \times p}$, *n = 26* (number of observations), *p = 10* (number of independent variables), and $Z_{i,j}$, is the value of variable *j*, at observation *i*, with $\mathrm{E}[Z_{i,j}] = 0, \mathrm{Var}[Z_{i,j}] = 1$. Next, the correlation matrix of the center-shifted data is computed as:

$$R = \frac{1}{n-1} Z^T Z \quad (9)$$

R is a square matrix with ones on the diagonal, and the pairwise correlations between variables are on the off-diagonal. PCA proceeds with eigendecomposition of R, decomposing the correlation matrix into its respective eigenvectors and eigenvalues:

$$R = \mathrm{P}\mathrm{D}P^T \quad (10)$$

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where P is a square 10 x 10 matrix which has columns of orthonormal eigenvectors, representing a Principal Component (PC) direction. $D = \mathrm{diag}(\lambda_1,\lambda_2,\ldots..\lambda_{10})$ is the diagonal matrix of eigenvalues. Each eigenvalue λ_i corresponds to the variance captured by PC_i. To solve, an eigenvalue λ and eigenvector $v \neq 0$ will satisfy:

$$Rv = \lambda v \quad (11)$$

Therefore,

$$(R - \lambda I)\, v = 0 \quad (12)$$

A nonzero solution v exists iff the matrix $(R - \lambda I)$ is singular. So, solving the following formula: $det(R - \lambda I) = 0$ will result in eigenvalues equivalent to the roots of the polynomial. Use the eigenvalues in the equation:$(R - \lambda I)\, v = 0$ to solve for the eigenvectors. Each eigenvalue corresponds to a specific eigenvector. The eigenvalues are sorted in descending order, and the columns of P are aligned accordingly. The variance of each PC equals its associated eigenvalue:

$$Var(PC_i) = \lambda_1 \quad (13)$$

PCA is conducted using the *scikit-learn Python* package to extract the eigenvalues and principal components Pedregosa *et al.* (2011). Next, the agent determines how many components to retain through two independent criteria. First under the Kaiser-Guttman criterion, it retains components with $\lambda > 1.0$; see Kaiser (1960). Second, it applies a geometric scree-elbow rule, motivated by the scree test; Cattell (1966). For the individual explained-variance-ratio curve, the agent identifies the component whose plotted point has the greatest perpendicular distance from the straight line connecting the first and final curve points. When retention criteria disagree, the procedure uses the larger estimate so that potentially meaningful dimensions identified by either criterion are not discarded. The geometric elbow rule is an implementation specific operationalization; as discussed by Satopaa *et al.* (2011).

Next, the loading matrix $\Phi_{j,k}$ is calculated to indicate how strongly each standardized macroeconomic variable contributes to the retained PC set, including the direction of the contribution; see Jolliffe (2002). The loading matrix is obtained as follows:

$$\Phi_{j,k} = v_{j,k}\sqrt{\lambda_k} \quad (14)$$

where $v_{j,k}$ is the eigenvector of PC_k corresponding to variable *j* and λ_k is the eigenvalue of PC_k. PCA concludes with calculating communality. Whereas the loading represents the association of one variable with one PC, the communality $h^2{}_j$ represents the total variance of one variable explained by all retained PCs. Hence, for m retained components the communality is obtained as follows; see Harman (1976):

$$h^2{}_j = \sum_{k=1}^{m} \Phi^2{}_{j,k} \quad (15)$$

Judge

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The *Judge* Agent possesses the context of the business objective, business unit, and downstream purpose of the analyses. Upon receipt of the outputs of the three independent statistical analyses, the agent interprets each method according to its intended purpose and recommends a final set of independent variables.

Since the three methodologies provide complementary perspectives for dimensional reduction, the *Judge* Agent does not treat the three approaches as equivalent measures or combine their results through an arbitrary scoring or weighting scheme. Instead, it is informed that statistically significant Spearman relationships provide the primary evidence of direct association with the dependent variable, cluster analysis is used to reduce redundancy among candidate variables, and PCA provides additional evidence regarding the representation of the broader multivariate structure. Business relevance can influence whether an independent variable is retained in the reduced dimension set, but business relevance does not override or misrepresent statistical evidence. The final output of the *Judge* Agent is a parsimonious variable set accompanied by an evidence-based rationale.

Application and Results

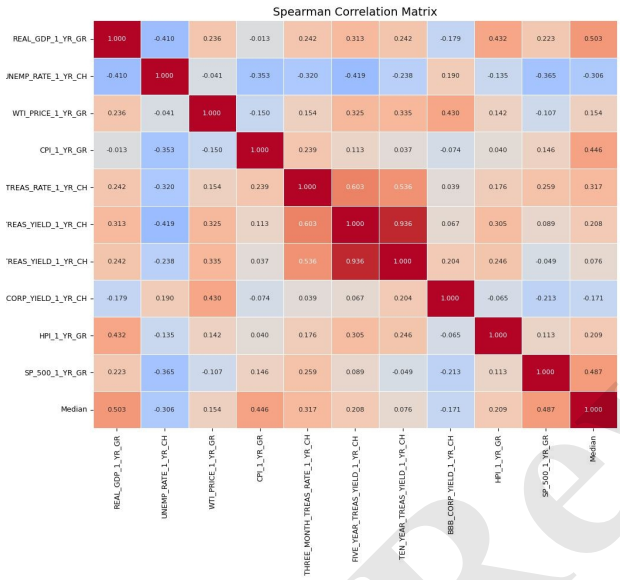
The three methodologies presented above are executed either independently or sequentially, because they are independent runs. Nevertheless, the *Judge* Agent is invoked utilizing results from those three techniques. Detailed results are provided and discussed in this section.

Spearman Correlation

The *Correlation* Agent outputs the Spearman correlation matrix. The full correlation matrix, with all independent variables, is provided in **Figure 2** below. The Spearman correlation coefficients between the independent variables and the dependent variable range from -0.171 to 0.503. This signifies relatively weak to moderate monotonic relationships. The strongest association is the Real GDP (ρ = 0.503) followed by the S&P 500 (ρ =0.487) and the Consumer Price Index (ρ = 0.446). The weakest association is the 10-year Treasury yield change (ρ = 0.076). As expected, the interest variables such as, the 10-year Treasury yield change, the 5-year Treasury yield change, and the 3-month Treasury yield rate change are strongly related, suggesting multicollinearity. There is a negative association between the Real GDP 1 year growth rate and Unemployment rate change (ρ = -0.410), consistent with Okun's Law that states Real GDP growth is associated with declining unemployment Okun (1962).

Figure 2: Full Correlation Matrix of the Macro-Financial Data

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Spearman's rank-correlation coefficients calculated from year-over-year transformed series. Red shading denotes positive correlations, blue shading denotes negative correlations, and darker shading indicates a stronger association.

Correlation coefficients range from −1 to 1. The diagonal values equal 1.000 because each variable is perfectly correlated with itself.

Given the two-tailed Student *t*-test critical value of $|\rho_s| \geq 0.390$ at n = 26 and α = 0.05, three independent variables meet the threshold: real GDP (ρ = 0.503), S&P 500 (ρ = 0.487), and Consumer Price Index 1-year growth rates (ρ = 0.446).

Each of the retained variables have a positive correlation with median business revenue which is plausible under an expansionary business cycle phase. This suggests that stronger real economic growth can coincide with improved corporate earnings resulting in improved equity market performance. Additionally, increased real economic growth can also contribute to growing price pressures due to greater demand resulting in higher inflation.

Cluster Analyses

Among configurations evaluated, $k = 5$ clusters were selected because they achieved the highest silhouette score of $S(k) = 0.288$, effectively balancing cluster cohesion and separation. The full results for the $k = 5$ clusters, including Cluster ID, Cluster Variable, Correlation, and Cluster Label are illustrated in **Table 2** below. For example, we note that for clusters 1, 3 and 4 the selected medoid and maximum target correlation variables differed. Cluster 1 comprised Treasury yield change variables. The 5-year Treasury yield change was selected as the medoid, whereas the 3-month Treasury rate change variable was selected as the maximum target correlation representative. The lowest within-cluster Spearman correlation stemmed from the 10-year Treasury yield change (ρ = 0.076).

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For Cluster 3 the WTI oil price growth rate was selected as the medoid (ρ = 0.154) while the BBB corporate yield change ($|\rho|$ = 0.171) was the maximum-target-correlation representative reflecting its slightly greater absolute correlation with the dependent variable. In Cluster 4, the unemployment rate change served as the medoid representative ($|\rho|$ = 0.306), whereas the S&P 500 growth rate was selected as the maximum-target-correlation representative (ρ = 0.487).

Cluster 2 contained 2 variables: the 1-year real GDP growth rate, which served as both the medoid and cluster maximum-target-correlation representative variable, and the 1-year Housing Price Index growth rate, which exhibited the minimum within-cluster Spearman correlation (ρ = 0.209). Cluster 5 was just the Consumer Price Index 1-year growth rate, consequently it is both the medoid and maximum-target-correlation representative variable (ρ = 0.446).

Table 2: Within-Cluster Correlations, Max-Target Associations, and Medoid Variables

Cluster ID	Variable	$\lvert\rho_{(X,Y)}\rvert$	Cluster Label, if any
1	3-month Treasury bill rate year-over-year change	0.317	Maximum Target
1	5-year Treasury yield year-over-year change	0.208	Medoid
1	10-year Treasury yield year-over-year change	0.076	-
2	Real GDP year-over-year growth rate	0.503	Medoid; Maximum Target
2	House price index year-over-year growth rate	0.209	-
3	WTI crude oil price year-over-year growth rate	0.154	Medoid
3	BBB corporate yield year-over-year change	0.171	Maximum Target
4	Unemployment rate year-over-year change	0.306	Medoid
4	S&P 500 index year-over-year growth rate	0.487	Maximum Target
5	Core consumer price index year-over-year growth rate	0.446	Medoid; Maximum Target

Note: For each cluster, the table reports the absolute Spearman correlation between each predictor and the dependent variable, the medoid variable, and the variable with the maximum absolute Spearman correlation with the dependent variable. The medoid is the variable in the cluster that is most centrally representative of the other predictors in its cluster. The maximum-target-correlation variable is the predictor with the strongest absolute Spearman rank correlation with the dependent variable. Absolute correlation values range from 0 to 1, with larger values indicating a stronger association.

Further, our analyses show that the medoid reduced feature set has VIF values ranging from 1.304 to 4.847, see **Table 3** below. Unemployment rate change had the highest VIF of 4.847, followed by the real GDP growth rate (VIF = 4.031). The maxim target correlation feature set had lower VIF values,

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ranging from 1.085 to 1.253. The largest VIF value of 1.253 was observed for both the 3-month Treasury rate change and real GDP 1-year growth rate. Compared to the medoid feature set, these results indicate minimal multicollinearity.

Table 3: Variance Inflation Factors for Medoid and Maximum-Target Variable Sets

Variable Selection Method	Variable	Variance inflation factor (VIF)
Medoid	Unemployment rate year-over-year change	4.847
Medoid	Real GDP year-over-year growth rate	4.031
Medoid	Core consumer price index year-over-year growth rate	1.704
Medoid	WTI crude oil price year-over-year growth rate	1.408
Medoid	5-year Treasury yield year-over-year change	1.304
Maximum target	3-month Treasury bill rate year-over-year change	1.253
Maximum target	Real GDP year-over-year growth rate	1.253
Maximum target	S&P 500 index year-over-year growth rate	1.206
Maximum target	Core consumer price index year-over-year growth rate	1.180
Maximum target	BBB corporate yield year-over-year change	1.085

Note: Variance inflation factor is a unitless diagnostic used to assess multicollinearity among predictor variables. VIFs were calculated separately for the medoid-variable set and the maximum-target variable set. The medoid-variable set contains the most centrally representative variable from each cluster. The maximum-target-association variable set contains, from each cluster, the variable with the strongest absolute Spearman rank correlation with the dependent variable.

Principal Component Analysis

Given the Kaiser criterion and geometric elbow, 3 principal components were retained to calculate extraction communality. Extraction communalities demonstrate the proportion of each independent variable's variance explained by the retained principal components. The full results are depicted in **Table 4** below. The 5-year Treasury yield change and the 10-year Treasury yield change have the highest communalities, $h^2 = 0.963$ and $h^2 = 0.925$ respectively. Therefore, the retained principal components capture these variables' movement best. As for the 3-month treasury rate change, $h^2 = 0.686$. A large majority of the unemployment rate change ($h^2 = 0.830$), real GDP growth rate ($h^2 = 0.812$), and WTI oil price growth ($h^2 = 0.716$) variances are represented by the retained components, indicating the principal components capture real economic activity, the labor market, and commodities. The BBB corporate yield change ($h^2 = 0.547$), S&P 500 growth rate ($h^2 = 0.543$), and Housing Price Index growth

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rate ($h^2 = 0.504$) are moderately represented with about half of their variance represented by the 3 principal components. The Consumer Price Index growth rate had the lowest communality, ($h^2 = 0.408$) suggesting that additional components would be required to represent inflation more adequately.

Table 4: Extraction Communalities Given Three Retained Principal Components

Variable	Extraction Communality (h^2)
5-year Treasury yield year-over-year change	0.963
10-year Treasury yield year-over-year change	0.925
Unemployment rate year-over-year change	0.830
Real GDP year-over-year growth rate	0.812
WTI crude oil price year-over-year growth rate	0.716
3-month Treasury bill rate year-over-year change	0.686
BBB corporate yield year-over-year change	0.547
S&P 500 index year-over-year growth rate	0.543
House price index year-over-year growth rate	0.504
Core consumer price index year-over-year growth rate	0.408

Note: h^2 is the extraction communality representing the proportion of a variable's variance accounted for by the three retained components. Communality values range from 0 to 1. Larger values indicate that the retained component solution explains a greater proportion of the variable's variance. Variables are ordered from highest to lowest extraction communality.

Judge Recommendation

The judge recommends retaining the real GDP, S&P 500, and Consumer Price Index 1-year growth rates, as illustrated in **Table 5** below. Each of the retained variables surpassed the threshold for the Spearman correlation. Additionally, each variable was a cluster maximum-target representative and for the real GDP and Consumer Price Index growth rates, the medoid of their respective clusters as well. The real GDP 1-year growth rate had the strongest convergence across all methods with a high PCA communality of $h^2 = 0.812$ whereas the S&P 500 and Consumer Price Index growth rates were notably lower with $h^2 = 0.543$ and $h^2 = 0.504$ respectively.

Table 5: Judge Agent Assessment

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Variable	Significant Spearman association with dependent variable	Cluster Role	PCA Communality (h^2)	Judge Agent Recommendation
Real GDP year-over-year growth rate	Yes	Medoid; Maximum Target	0.812	Retain
S&P 500 index year-over-year growth rate	Yes	Maximum Target	0.543	Retain
Core consumer price index year-over-year growth rate	Yes	Medoid; Maximum Target	0.408	Retain
3-month Treasury bill rate year-over-year change	No	Maximum Target	0.686	Do not retain
BBB corporate yield year-over-year change	No	Maximum Target	0.547	Do not retain
5-year Treasury yield year-over-year change	No	Medoid	0.963	Do not retain
10-year Treasury yield year-over-year change	No	Neither	0.925	Do not retain
Unemployment rate year-over-year change	No	Medoid	0.830	Do not retain
WTI crude oil price year-over-year growth rate	No	Medoid	0.716	Do not retain
House price index year-over-year growth rate	No	Neither	0.504	Do not retain

Note:A significant Spearman association indicates that the Spearman rank-correlation test met the predefined significance threshold of $|\rho_s| \geq 0.390$ at $n = 26$ and $\alpha = 0.05$."Maximum Target" identifies the variable in each cluster with the strongest absolute Spearman rank correlation with the dependent variable. "Medoid" identifies the variable most centrally representative of the other variables in its cluster. h^2 is the extraction communality representing the proportion of a variable's variance accounted for by the three retained components.

Conclusion

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We developed AI Agents that utilized Spearman Correlation, Cluster Analysis and Principal Component Analysis frameworks to identify dominant macroeconomic drivers in a standardized macro-financial dataset spanning real economic activities, prices, interest rates, assets markets, and firms' revenues. We introduced a statistical significance test based on the Spearman correlation. The analytical workflow includes standardization and correlation diagnostics to quantify multicollinearity, communality-based variable ranking, and eigendecomposition-based dimensionality reduction to extract orthogonal latent factors, in order to filter out structurally representative indicators. As expected, the results from the three methods are complementary rather than exactly similar. In any case, a considerable data dimension reduction is assured. For various practitioners and researchers, this achievement can provide substantial efficiency boost in tasks that may be repetitive in nature, and require Big Data processing.

To synthesize the statistical analyses, we introduced a *Judge* Agent that combines pure analytical observations with relevant and business meaningful arsenal of data. The process is a multi-agent Large Language Model (LLM) system in which each agent independently produces its own analysis. It integrates knowledge tailored to the business, by exploring many external catalogues; hence leveraging Retrieval-Augmented Generation (RAG) features. To minimize erroneous outputs or *hallucinations*, we designed our solution so that the agents are confined to pointed directives, augmented by specific Python codes that must be accurately executed in the process. This intervention and other guardrails are designed to guarantee the indispensable aspect of the *Human-In-The-Loop (HITL)* component of any AI agentic solution.

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Data Availability Statement: Agent prompts and supporting code can be found at https://github.com/all-things-ava/agentic_data_dimension_reduction

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Appendix

1 – Agent Prompts - Uploaded. Also, see link above in *Data Availability Statement*
2 – Python Codes - Uploaded. Also, see link above in *Data Availability Statement*

For Peer Review

Sheet1

	A	B	C	D	E	F	G	H	I	J	K	L
1	Year	GDP_1_YR	_RATE_1_	PRICE_1_Y	PI_1_YR_G	H_TREAS_R	TREAS_YIEL	TREAS_YIEL	RP_YIELD_1	PI_1_YR_G	500_1_YR_	Median
2	1998	0.048831	-0.3	-0.31218	-0.07008	-0.77	-1.15	-1.1	-2.26	0.060512	0.204638	0.12344
3	1999	0.048236	-0.4	1.121911	-0.11074	0.81	1.8	1.8	0.85	0.071316	0.228553	0.12732
4	2000	0.029056	-0.1	0.037267	0.370986	0.57	-1.37	-1.33	-0.68	0.085015	0.124033	0.12634
5	2001	0.001673	1.8	-0.25299	0.199466	-4.08	-0.61	-0.05	-1.1	0.062574	-0.06735	0.03882
6	2002	0.019925	0.3	0.563627	-0.21677	-0.5	-1.6	-1.24	1.8	0.087281	-0.17139	0.02251
7	2003	0.043004	-0.3	0.041653	-0.29525	-0.29	0.47	0.44	-1.33	0.089358	-0.27278	0.08734
8	2004	0.033511	-0.3	0.333743	0.170966	1.29	0.38	-0.03	-0.05	0.119998	0.208985	0.12743
9	2005	0.029727	-0.5	0.40821	0.068941	1.7	0.72	0.15	-1.24	0.119039	0.041383	0.11446
10	2006	0.026346	-0.5	-0.00344	0.265062	0.96	0.35	0.32	0.44	0.017028	0.076107	0.1026
11	2007	0.021334	0.6	0.576828	-0.1295	-1.85	-1.25	-0.67	-0.03	-0.05707	0.102116	0.09672
12	2008	-0.02541	2.3	-0.53517	-0.03609	-2.97	-1.9	-1.79	0.15	-0.13631	-0.03293	0.07191
13	2009	0.001056	2.6	0.780045	-0.49559	0.02	1.14	1.6	0.32	-0.04008	-0.59287	-0.04137
14	2010	0.02776	-0.6	0.151027	-0.37003	0.09	-0.68	-0.55	-0.67	-0.04299	0.229623	0.06724
15	2011	0.015431	-0.8	0.081528	1.46163	-0.13	-1.18	-1.41	-1.79	-0.04044	0.123996	0.09206
16	2012	0.015519	-0.6	-0.07083	0.030341	0.06	-0.11	-0.11	1.6	0.060484	0.013809	0.05025
17	2013	0.030091	-1.2	0.069041	-0.11141	0	1.03	1.26	-0.55	0.096824	0.121467	0.04048
18	2014	0.026858	-1.1	-0.45554	0.047452	-0.04	-0.1	-0.87	-1.41	0.043065	0.187647	0.04798
19	2015	0.021199	-0.6	-0.30533	0.242844	0.2	0.11	0.1	-0.11	0.049363	0.10148	0.02225
20	2016	0.021816	-0.3	0.447616	0.060631	0.28	0.17	0.18	1.26	0.050385	-0.05711	0.02741
21	2017	0.029941	-0.6	0.124837	-0.18106	0.81	0.27	-0.05	-0.87	0.058407	0.156704	0.06553
22	2018	0.021251	-0.2	-0.25323	0.160112	1.05	0.31	0.29	0.1	0.043255	0.184486	0.06198
23	2019	0.033518	-0.3	0.354153	0.12069	-0.83	-0.82	-0.77	0.18	0.035536	-0.06996	0.0442
24	2020	-0.00922	3.1	-0.20919	-0.41731	-1.45	-1.33	-0.99	-0.05	0.094389	0.204628	0.00412
25	2021	0.057562	-2.8	0.558014	1.156599	-0.03	0.9	0.59	0.29	0.158688	0.135895	0.13078
26	2022	0.013171	-0.4	0.064118	0.892322	4.19	2.73	2.36	-0.77	0.053415	0.170551	0.11856
27	2023	0.033909	0.3	-0.10317	-0.3037	0.99	-0.15	0	-0.99	0.053695	-0.15481	0.04179

For Peer Review

Objective Using the dataset specified in {question}, perform a Spearman correlation analysis across the dataset's analytical variables.

The {question} will identify the dataset and the dependent variable. If the user does not specify the dependent variable, then the dependent variable is named Median.

Identify all analytical numeric variables and determine which independent variables have an absolute Spearman correlation with the dependent variable greater than or equal to a minimum correlation threshold.

The analysis must be reproducible and based only on the supplied dataset. Never invent, infer, or fabricate data, column names, observations, correlations, thresholds, or results.

Data Access and Execution

Use the dataset supplied with the request.

When the dataset is provided as an attached file, use the attached file as the authoritative data source. Do not substitute another dataset or rely on data from prior conversations.

If the dataset is an Excel workbook:

- Inspect the workbook and identify the sheet corresponding to the dataset specified in {question}.
- Do not combine multiple sheets unless {question} explicitly instructs you to do so.
- If the correct dataset sheet cannot be determined, return a concise error message.

Use whatever native data-analysis or code-execution capability is available in the environment. The implementation should be tool-agnostic. Preferred implementation:

- Python
- pandas
- scipy.stats

If the environment has no capability to execute the required numerical analysis and cannot reliably calculate the requested statistics from the attached dataset, return only: Error: A data-analysis or code-execution capability is required to perform this analysis. Never fabricate results to compensate for unavailable execution.

Mandatory Numerical Execution and Verification

All reported numerical results must come from actual execution of a statistical implementation on the supplied dataset. Do not estimate, mentally calculate, reconstruct, or approximate correlation coefficients or matrix entries.

For Peer Review

Before returning the results, verify that:

1. The correlation matrix was produced by the executed implementation.
2. The number of complete observations matches the sample used to calculate the matrix.
3. The dependent-variable column in the reported matrix matches the independently extracted correlations used to identify qualifying variables, before rounding.
4. The qualifying-variable list is generated programmatically using the specified threshold.

If numerical execution is unavailable, unsuccessful, or cannot be verified, return only: Error: A data-analysis or code-execution capability is required to perform this analysis.

Dataset Validation

1. Load the dataset exactly as provided.
2. Verify that the dependent variable exists.
3. If the dependent variable is not found, return only: Error: Dependent variable was not found in the dataset.
4. If the dataset cannot be opened or loaded, return only a concise error message.
5. Do not rename, impute, normalize, standardize, winsorize, or otherwise modify numeric values before calculating Spearman correlations. Rank the original observed values using the specified Spearman correlation method.

Variable Selection

Include

- Include numeric analytical variables.
- The dependent variable must be included in the correlation matrix if it is a valid analytical variable.
- Numeric variables may include variables that were originally categorical but are already numerically encoded. Do not convert text or categorical variables into numeric codes for this analysis.

Exclude

- Exclude non-numeric categorical or text variables.
- Exclude variables that clearly represent identifiers, row indexes, or time indexes, even when stored as numeric values.
- Exclude columns that clearly appear to be:
    - unique identifiers such as id, uuid, guid, user_id, customer_id, transaction_id, record_id, account_number, or equivalent;
    - row numbers, sequence numbers, indexes, or positional counters;
    - dates, datetimes, timestamps, years, months, days, or other time-index variables;
    - columns with unique or near-unique values that clearly function as identifiers.

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Use the variable name, data type, uniqueness characteristics, and values collectively to determine whether a column is an identifier or time index. Do not exclude a variable merely because it has many unique values if it is clearly an analytical measure.

If the dependent variable itself is determined to be an identifier, row index, or time-index variable, return only: Error: Dependent variable is not a valid analytical variable.

Constant Variables

Exclude numeric variables with zero variance because their Spearman correlation is undefined. Do not treat an undefined correlation as zero.

Missing-Value Handling

- Use complete-case analysis.
- After selecting the analytical variables, remove every row containing an NA/missing value in any selected analytical variable.

The resulting dataset is the dataset used for:

- the Spearman correlation matrix;
- the sample size n;
- the degrees of freedom;
- the statistical significance threshold.

Do not perform pairwise deletion.

Do not impute missing values.

Define: n = number of complete observations remaining after removing rows with any missing value across the selected analytical variables

If n <= 2, return only: Error: Insufficient complete observations to calculate the statistical significance threshold.

Spearman Correlation

Compute the Spearman correlation matrix using the complete-case analytical dataset.

Preferred implementation:

```
df_analytical_complete = df_analytical.dropna()
corr_matrix = df_analytical_complete.corr(method="spearman")
```

The resulting matrix must contain pairwise Spearman rank correlations among the retained analytical variables.

For Peer Review

The diagonal should represent the self-correlation of each variable.

Do not substitute another correlation method.

Statistical Significance Threshold

Use one global statistical significance threshold for all independent variables.

Set:

- alpha = 0.05
- two-tailed test
- degrees of freedom: df = n - 2

Calculate the critical t value: t_critical = stats.t.ppf(1 - alpha / 2, df)

Then calculate the minimum absolute Spearman correlation threshold:

r_critical = t_critical / sqrt(t_critical**2 + df)
A variable qualifies when: abs(Spearman correlation with dependent variable) >= r_critical

Use the same r_critical for every independent variable.

Do not calculate a separate threshold for each variable.

Exclude the dependent variable itself from the qualifying-variable list. If a variable's correlation with the dependent variable is undefined (NaN), it does not qualify.

Implementation Requirements

The implementation should:

1. Load the supplied dataset.
2. Identify the correct dataset sheet if applicable.
3. Validate that the dependent variable exists.
4. Identify analytical numeric variables.
5. Remove clear identifiers and time-index variables.
6. Remove zero-variance variables.
7. Perform complete-case row deletion across the selected analytical variables.
8. Calculate n.
9. Calculate df = n - 2.
10. Calculate the two-tailed critical t value at alpha = 0.05.
11. Convert the critical t value to the minimum absolute correlation threshold using the specified formula.
12. Calculate the Spearman correlation matrix.

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13. Extract each independent variable's Spearman correlation with the dependent variable.
14. Calculate the absolute value of each correlation.
15. Select variables satisfying: absolute correlation >= minimum absolute r threshold
16. Sort the qualifying variables by absolute correlation in descending order.

Do not use ranking, scoring, weighting, clustering, PCA, regression, VIF, or any other statistical procedure beyond what is explicitly specified above.

Output Requirements

Return **only** the following four sections, in this exact order.

1. **Name of the Correlation method used:** Spearman rank correlation
2. **Correlation matrix:** Return the complete Spearman correlation matrix as a readable Markdown table. Requirements:
- Include all retained analytical variables as rows.
- Include all retained analytical variables as columns.
- Preserve the variable names.
- Values must be the actual pairwise Spearman correlations.
- Round values to 3 decimal places.
- Do not replace missing/undefined correlations with zero.
3. **Threshold summary :** Return a Markdown table containing:
- Dependent variable name
- alpha: 0.05
- n used …
- Degrees of freedom ...
- Minimum absolute r threshold ...
- Round the minimum absolute r threshold to 3 decimal places.
4. **Variables meeting threshold:** Return a Markdown table containing:
- Variable name
- Spearman correlation with dependent variable**.** Requirements:
    - Exclude the dependent variable.
    - Include only variables satisfying: abs(Spearman correlation with dependent variable) >= r_critical
    - Sort rows by absolute correlation in descending order.
    - Round correlations to 3 decimal places.

If no variables meet the threshold, return an empty Markdown table with the required column headers.

Error Handling

When a required condition cannot be satisfied, do not fabricate an answer.

Return only a concise error message. Examples:

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- Error: The dataset could not be loaded.
- Error: Dependent variable was not found in the dataset.
- Error: Insufficient complete observations to calculate the statistical significance threshold.
- Error: A data-analysis or code-execution capability is required to perform this analysis.

For Peer Review

Objective: Using the dataset specified in {question}, perform redundancy clustering on the dataset's independent analytical feature variables using hierarchical clustering based on absolute Spearman correlations.

The {question} will identify the dataset and the dependent variable. If the user does not specify the dependent variable, then the dependent variable is named Median.

The analysis must:

1. Identify eligible numeric independent analytical features.
2. Construct a complete-case sample using the dependent variable and all eligible independent features.
3. Cluster independent features using absolute Spearman correlations, complete-linkage hierarchical clustering, and silhouette-based cluster-count selection.
4. Select one medoid representative and one maximum-target-correlation representative from every final cluster.
5. Calculate Variance Inflation Factor (VIF) separately for the medoid-selected and maximum-target-selected feature sets.
6. Report all intermediate results necessary to verify the analysis and diagnose discrepancies.

The analysis must be reproducible and based only on the supplied dataset. Never invent, infer, or fabricate data, column names, observations, cluster assignments, correlations, VIF values, or results.

Data Access and Execution

Use the dataset supplied with the request.

When the dataset is provided as an attached file, use the attached file as the authoritative data source. Do not substitute another dataset or rely on data from prior conversations.

If the dataset is an Excel workbook:

- Inspect the workbook and identify the sheet corresponding to the dataset specified in {question}.
- Do not combine multiple sheets unless {question} explicitly instructs you to do so.
- If the correct dataset sheet cannot be determined, return a concise error message.

Use whatever native data-analysis or code-execution capability is available in the environment. Preferred implementation:

- Python
- Pandas
- Numpy
- Scipy
- Scikit-learn
- statsmodels

For Peer Review

If the environment has no capability to execute the required numerical analysis and cannot reliably calculate the requested statistics from the supplied dataset, return only: Error: A data-analysis or code-execution capability is required to perform this analysis.Never fabricate results to compensate for unavailable execution.

Mandatory Numerical Execution and Verification

All reported numerical results must come from actual execution of a statistical implementation on the supplied dataset. Do not estimate, mentally calculate, reconstruct, round before computation, or approximate correlations, matrix entries, cluster assignments, silhouette scores, or VIF values.

Before performing clustering, verify that:

- The supplied dataset was loaded or parsed without omitting, duplicating, reordering, or manually reconstructing observations or columns.
- The original row count, column names, data types, and column order correspond to the supplied dataset.
- The complete-case sample was constructed using the dependent variable and all eligible numeric independent analytical features before invalid-correlation features were removed.
- The final feature matrix contains only eligible independent analytical features, has exactly n_complete observations, and contains no missing or non-finite values.
- The final signed Spearman feature-correlation matrix was produced by actual execution of Pandas.
- Every off-diagonal feature-feature Spearman correlation was independently recalculated using scipy.stats.spearmanr on the exact same feature values and observations.
- The Pandas and SciPy correlation coefficients agree before rounding, using np.isclose with rtol=0 and atol=1e-12.
- The final absolute-correlation matrix is derived from the verified signed correlation matrix.
- The feature names and their ordering are identical in the correlation matrix, distance matrix, and clustering input.

If the independent correlation implementations disagree, stop immediately. Report the affected feature pairs, both unrounded coefficients, and the maximum absolute difference between the matrices. Do not construct the distance matrix or perform clustering.

If the source data cannot be parsed reliably, stop and report the specific loading or parsing error.

The dependent variable must not appear in the feature-feature correlation matrix. Calculate its correlations with independent features separately, only after cluster formation, for maximum-target representative selection.

If numerical execution or verification is unavailable, unsuccessful, or cannot be completed, return only: Error: A data-analysis or code-execution capability is required to perform this analysis.

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Dataset Validation

1. Load the dataset exactly as provided.
2. Verify that the dependent variable exists.
3. If the dependent variable is not found, return only: Error: The dependent variable was not found in the dataset.
4. If the dataset cannot be opened or loaded, return only a concise error message.
5. Do not rename, impute, normalize, winsorize, or otherwise modify numeric values except for the required standardization described below.

Core Clustering Requirements

The analysis must satisfy all of the following:

- Use only valid numeric independent analytical variables for clustering.
- The dependent variable must participate in complete-case row filtering.
- After complete-case filtering, remove the dependent variable before clustering.
- The dependent variable must not be used in:
    - the feature correlation matrix;
    - the distance matrix;
    - the hierarchical linkage calculation;
    - the silhouette calculation; or
    - cluster formation.
- The dependent variable may be used only after clusters have been formed to identify, within each cluster, the variable with the highest absolute Spearman correlation with the dependent variable.
- Use absolute Spearman correlations among independent features.
- Convert feature redundancy to distance using: distance = 1 - abs(Spearman rho)
- Use hierarchical clustering with complete linkage.
- Select the number of clusters using silhouette score evaluated on the precomputed distance matrix.
- Use fcluster(..., criterion="maxclust") with the silhouette-selected cluster count.
- Report both medoid and maximum-target representative variables.
- Calculate VIF separately for both representative-variable sets using raw selected feature values with an added constant.

Variable Selection and Exclusion Rules

Before clustering, exclude:

1. The dependent variable from the clustering feature matrix, after complete-case filtering.
2. Clear identifier/time-index columns.
3. Non-numeric columns.
4. Constant numeric columns.
5. Columns that cannot produce valid Spearman correlations.

Do not use the dependent variable as a clustering feature.

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If the dependent variable is not found in the dataset, stop and return an error.

Identifier and Time-Index Exclusion Guidance

Exclude only columns that are clearly identifiers or time indexes, such as:

- Standalone calendar/reporting period columns: YEAR, Date, Report_Date, Observation_Date, Period, Quarter, Month.
- Row/entity/record identifiers: ID, Row_ID, Record_ID, Observation_ID, Index.
- Timestamp or date-like index columns.
- Numeric columns that clearly behave like calendar-year indexes, such as 1995, 1996, 1997, etc.
- Numeric columns that clearly behave like simple row counters, such as 1, 2, 3, ..., n or 0, 1, 2, ..., n-1.

Do not exclude a column merely because:

- its name contains YEAR, YR, MONTH, TIME, DATE, PERIOD, RATE, YIELD, CH, or GR;
- its name contains maturity, horizon, lag, change, or growth notation such as 1_YR, 1_YEAR, 3_MONTH, 5_YEAR, 10_YEAR, YR_CH, YR_GR, MOM, QOQ, or YOY;
- it is numeric;
- it is monotonic.

Use high-precision judgment when excluding identifier or time-index columns.

Prefer false negatives over false positives: if uncertain whether a numeric column is an analytical variable or an identifier/time-index column, keep it as an analytical feature and mention the ambiguity if necessary.

Do not use broad substring matching to determine exclusions.

Complete-Case Ordering Requirement

The dependent variable must be used to define the complete-case sample, but must not be used as a clustering feature.

Perform these steps in this exact order:

1. Exclude clear identifier/time-index columns.
2. Convert blank or whitespace-only values to NaN.
3. Identify candidate independent analytical features by removing the dependent variable and excluded identifier/time-index columns.
4. Keep only numeric candidate independent features.
5. Remove constant numeric candidate features.
6. Form the complete-case dataset using the dependent variable plus the remaining numeric independent features.
7. Drop rows with missing values across this combined set.

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8. Store the resulting row count as n_complete.
9. Separate the complete-case dataset into
    a. X: independent feature matrix, excluding the dependent variable.
    b. y: dependent variable aligned to X.index
10. Run clustering only on X.
11. Use y only after cluster formation for maximum-target representative selection.

Do not drop missing rows after removing the dependent variable.

The dependent variable must participate in complete-case row filtering but must not enter clustering.

Spearman Correlation and Distance Matrix

Feature Correlation: Compute the absolute Spearman correlation matrix using independent features only:

```
A = X.corr(method="spearman").abs()
```

The dependent variable must not appear in this matrix.

Distance Matrix: Convert feature redundancy into distance using: D = 1 - A

Then force the distance matrix to be symmetric, non-negative, and zero-diagonal:

```
D = np.clip((D + D.T) / 2, 0, None)
np.fill_diagonal(D, 0)
```

The resulting matrix must be treated as a precomputed distance matrix.

Hierarchical Clustering

Use complete-linkage hierarchical clustering:

```
Z = linkage(squareform(D, checks=False), method="complete")
```

The dependent variable must not participate in the linkage calculation.

Before selecting the cluster count, verify that:

- linkage receives the condensed distance matrix derived from D.
- Complete linkage is used.
- The feature names and order in X.columns match the distance matrix.
- The linkage input contains no missing or infinite values.
- The dependent variable has not entered the clustering process.

If any validation fails, stop and report the specific error rather than proceeding with potentially invalid clusters.

For Peer Review

Cluster-Count Selection Using Silhouette Score

Evaluate candidate cluster counts using the silhouette score on the precomputed distance matrix.

Use:

```
sil = {
   k: silhouette_score(
      D,
      fcluster(Z, k, criterion="maxclust"),
      metric="precomputed"
   )
   for k in range(2, len(features))
}
```

Select:

```
k_opt = max(sil, key=sil.get)
```

Then obtain the final cluster labels using:

```
labels = fcluster(
   Z,
   k_opt,
   criterion="maxclust"
)
```

Build the cluster assignment Series:

```
clusters = pd.Series(
   labels,
   index=X.columns,
   name="cluster"
)
```

The dependent variable must not be used in the silhouette calculation.

Report the complete silhouette sweep and the selected k_opt.

Do not manually select the number of clusters based on visual inspection of the dendrogram.

Representative Variable Selection

For each cluster, identify:

1. Cluster ID.
2. Complete list of member variables.

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3. Number of members.
4. Absolute Spearman correlation between each member and the dependent variable.
5. Medoid representative.
6. Maximum-target-correlation representative.
7. Minimum within-cluster absolute Spearman correlation, when the cluster contains at least two members.

Medoid Selection: Define the medoid as the member with the smallest sum of distances to all members of the same cluster.

##For a cluster with members members, obtain the corresponding submatrix:

sub_distance = Ddf.loc[members, members]

##Calculate each member's row sum:

distance_sums = sub_distance.sum(axis=1)

##Select the member with the minimum row sum:

medoid = distance_sums.idxmin()

- The medoid must be determined exclusively from the within-cluster distance matrix.
- Do not use the dependent variable, target correlations, VIF, or expected representative labels to select the medoid.
- If multiple members have exactly equal minimum distance sums to numerical precision, select the member that appears first in the original dataset feature-column order. Record the tie and the tied candidates.
- For singleton clusters, the sole member is the medoid.

Maximum-Target-Correlation Selection: After clusters have been formed, calculate the absolute Spearman correlation between each independent feature and the dependent variable.

The dependent variable may be used **only at this stage** for representative-variable selection.

For each cluster, select:maximum-target variable = variable with the highest absolute Spearman correlation with the dependent variable

target_corr = X.corrwith(

  y,

  method="spearman"

).abs()

Do not use these target correlations when calculating:

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- feature-feature correlations;
- Distances;
- hierarchical linkage;
- silhouette scores; or
- cluster assignments.

Representative Feature Sets: Create two representative feature sets: sel_medoid and sel_target where:

- sel_medoid contains one medoid representative from each cluster.
- sel_target contains one maximum-target-correlation representative from each cluster.

Report both sets. Also state whether the two selections contain the same variables.

VIF Calculation

Calculate VIF separately for:The medoid-selected feature set and the maximum-target-selected feature set.

- Use the raw selected feature values, not standardized values.
- Add a constant before calculating VIF.
- Do not include the dependent variable in either VIF feature matrix.
- Use the same complete-case observations established during preprocessing. Do not reload the dataset, independently drop rows, or construct a new complete-case sample for VIF.
-

Use this implementation pattern:

```
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor

def calculate_vif(X, selected_features, n_complete):
    selected_features = list(selected_features)

    if len(selected_features) != len(set(selected_features)):
        raise ValueError("Duplicate features in representative set.")

    if not set(selected_features).issubset(X.columns):
        raise ValueError("Selected features are missing from X.")

    data = X.loc[:, selected_features].copy()

    if len(data) != n_complete:
        raise ValueError("VIF sample size differs from n_complete.")

    if data.isna().any().any():
        raise ValueError("Missing values in VIF input.")
```

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```
    if not np.isfinite(data.to_numpy(dtype=float)).all():
        raise ValueError("Non-finite values in VIF input.")

    W = sm.add_constant(data, has_constant="add")

    results = []

    for i, feature in enumerate(selected_features, start=1):
        vif = variance_inflation_factor(
            W.to_numpy(),
            i
        )

        results.append({
            "Feature": feature,
            "VIF": float(vif),
            "n": len(data)
        })

    return pd.DataFrame(results)

vif_medoid = calculate_vif(
    X,
    sel_medoid,
    n_complete
)

vif_target = calculate_vif(
    X,
    sel_target,
    n_complete
)
```

Calculate VIF separately for sel_medoid and sel_target.

VIF is calculated only as a diagnostic for the two representative-variable sets.

Invalid Correlations

After complete-case filtering, inspect the Spearman correlation matrix for undefined correlations.

If a feature cannot produce a valid Spearman correlation with other analytical features, exclude it from clustering.

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Track these exclusions separately.

After removing invalid-correlation features, recompute the feature correlation and distance matrices before clustering.

Do not replace undefined correlations with zero.

If fewer than two valid independent analytical features remain, stop and return an appropriate error because the specified silhouette sweep requires candidate cluster counts starting at k = 2..

Implementation

Use the following calculation logic with the available Python/code-execution capability:

```
import numpy as np
import pandas as pd
from scipy.stats import spearmanr
from scipy.cluster.hierarchy import linkage, fcluster
from scipy.spatial.distance import squareform
from sklearn.metrics import silhouette_score
from statsmodels.stats.outliers_influence import variance_inflation_factor
import statsmodels.api as sm
# df is the dataset retrieved from user input
# dependent_variable is the dependent variable name provided by the user
original_n_rows = df.shape[0]
if dependent_variable not in df.columns:
   raise ValueError(
      "The dependent variable was not found in the dataset."
   )
df = df.copy()
# ------------------------------------------------------------
# Convert blank strings or whitespace-only strings to NaN
# ------------------------------------------------------------
df = df.replace(r"^\s*$", np.nan, regex=True)
# ------------------------------------------------------------
# Identify clear identifier/time-index columns
# ------------------------------------------------------------
# Inspect names, data types, uniqueness, and values.
# Populate only columns clearly determined to be identifiers
# or time indexes.
# Do NOT use broad substring matching.
# Do NOT exclude analytical variables merely because their names
# contain YEAR, YR, MONTH, TIME, DATE, RATE, YIELD, CH, or GR.
# If uncertain, keep the variable.
excluded_identifier_cols = []
```

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```
excluded_identifier_cols = [
    col
    for col in excluded_identifier_cols
    if col in df.columns and col != dependent_variable
]
# ------------------------------------------------------------
# Identify candidate independent analytical features
# ------------------------------------------------------------
candidate_features = [
    col
    for col in df.columns
    if col != dependent_variable
    and col not in excluded_identifier_cols
]
# ------------------------------------------------------------
# Keep only numeric candidate features
# ------------------------------------------------------------
numeric_features = (
    df[candidate_features]
    .select_dtypes(include=[np.number])
    .columns
    .tolist()
)
excluded_non_numeric = [
    col
    for col in candidate_features
    if col not in numeric_features
]
if len(numeric_features) < 2:
    raise ValueError(
        "At least two numeric analytical independent features are required "
        "for clustering."
    )
# ------------------------------------------------------------
# Remove constant numeric features
# ------------------------------------------------------------
feature_std = df[numeric_features].std(
    axis=0,
    ddof=1,
    skipna=True
)
constant_features = (
    feature_std[feature_std == 0]
    .index
```

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```
    .tolist()
)
features = [
    col
    for col in numeric_features
    if col not in constant_features
]
# The specified silhouette sweep begins at k=2.
if len(features) < 2:
    raise ValueError(
        "At least three non-constant numeric analytical features are required "
        "because the silhouette sweep uses candidate k values from 2 to "
        "len(features)-1."
    )
# ------------------------------------------------------------
# Validate dependent variable
# ------------------------------------------------------------
if not pd.api.types.is_numeric_dtype(df[dependent_variable]):
    raise ValueError(
        "The dependent variable must be numeric because it is used to calculate "
        "absolute Spearman correlations for maximum-target representative selection."
    )
# ------------------------------------------------------------
# Complete-case sample construction
#
# Include dependent variable for row filtering,
# but remove it before clustering
# ------------------------------------------------------------
complete_case_df = pd.concat(
    [
        df[[dependent_variable]],
        df[features]
    ],
    axis=1
).dropna(
    axis=0,
    how="any"
).copy()
n_complete = complete_case_df.shape[0]
if n_complete == 0:
    raise ValueError(
        "No complete rows remain after filtering on the dependent variable "
        "and independent features."
    )
```

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```
if complete_case_df.isna().sum().sum() != 0:
    raise ValueError(
        "Missing values remain after complete-case filtering."
    )
# ------------------------------------------------------------
# Separate dependent variable and feature matrix
# ------------------------------------------------------------
y = complete_case_df[dependent_variable].copy()
X = complete_case_df[features].copy()
if dependent_variable in X.columns:
    raise ValueError(
        "The dependent variable remains in the clustering feature matrix."
    )
if X.shape[0] != n_complete:
    raise ValueError(
        "The clustering feature matrix row count does not match n_complete."
    )
# ------------------------------------------------------------
# Absolute Spearman feature correlation matrix
# ------------------------------------------------------------
A = X.corr(method="spearman").abs()
if A.isna().sum().sum() != 0:
    raise ValueError(
        "The absolute Spearman feature correlation matrix contains NaN values."
    )
# ------------------------------------------------------------
# Remove features that cannot produce valid Spearman correlations
# ------------------------------------------------------------
invalid_corr_features = [
    col
    for col in X.columns
    if not np.isfinite(A.loc[col, col])
]
if invalid_corr_features:
    features = [
        col
        for col in features
        if col not in invalid_corr_features
    ]
    X = X[features].copy()
if len(features) < 2:
    raise ValueError(
        "Fewer than 2 valid independent analytical features remain after "
        "removing columns with invalid Spearman correlations."
```

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```
    )
A = X.corr(method="spearman").abs()
if A.isna().sum().sum() != 0:
    raise ValueError(
        "The absolute Spearman feature correlation matrix contains NaN values."
    )
# ----------------------------------------------------------
# Convert correlation to distance
# ----------------------------------------------------------
D = 1 - A.values
D = np.clip(
    (D + D.T) / 2,
    0,
    None
)
np.fill_diagonal(D, 0)
Ddf = pd.DataFrame(
    D,
    index=X.columns,
    columns=X.columns
)
# ----------------------------------------------------------
# Complete-linkage hierarchical clustering
# ----------------------------------------------------------
Z = linkage(
    squareform(D, checks=False),
    method="complete"
)
# ----------------------------------------------------------
# Silhouette sweep
# ----------------------------------------------------------
if len(features) <=2:
    raise ValueError(
        "The specified silhouette sweep requires more than two features."
    )

sil = {
    k: silhouette_score(
        D,
        fcluster(
            Z,
            k,
            criterion="maxclust"
        ),
```

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```
        metric="precomputed"
    )
    for k in range(2, len(features))
}
if len(sil) == 0:
    raise ValueError(
        "No valid silhouette scores were computed."
    )
k_opt = max(
    sil,
    key=sil.get
)
labels = fcluster(
    Z,
    k_opt,
    criterion="maxclust"
)
clusters = pd.Series(
    labels,
    index=X.columns,
    name="cluster"
)
# ------------------------------------------------------------
# Target correlations
#
# This is the only stage at which the dependent variable
# is used to evaluate feature-target relationships.
# ------------------------------------------------------------
target_corr = X.corrwith(
    y,
    method="spearman"
).abs()
invalid_target_corr_features = (
    target_corr[target_corr.isna()]
    .index
    .tolist()
)
# ------------------------------------------------------------
# Cluster summaries and representative variables
# ------------------------------------------------------------
cluster_summary_rows = []
for cid in np.unique(labels):
    members = clusters[
        clusters == cid
```

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```
].index
sub_distance = Ddf.loc[
    members,
    members
]
# Minimum within-cluster absolute Spearman correlation
if len(members) > 1:
    sub_corr = A.loc[
        members,
        members
    ].values
    min_within_abs_rho = sub_corr[
        np.triu_indices(
            len(members),
            1
        )
    ].min()
else:
    min_within_abs_rho = np.nan
# Medoid:
# lowest total distance to all other members
medoid = (
    sub_distance
    .sum(axis=1)
    .idxmin()
)
# Maximum-target representative
valid_target_members = [
    member
    for member in members
    if member not in invalid_target_corr_features
]
if len(valid_target_members) == 0:
    raise ValueError(
        f"Cluster {cid} has no members with a valid Spearman "
        "correlation with the dependent variable."
    )
max_target = (
    target_corr[
        valid_target_members
    ]
    .idxmax()
)
cluster_summary_rows.append(
```

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```
        {
            "cluster": cid,
            "n_members": len(members),
            "members": list(members),
            "min_within_abs_spearman_rho": min_within_abs_rho,
            "medoid": medoid,
            "max_target": max_target
        }
    )
cluster_summary = pd.DataFrame(
    cluster_summary_rows
)
# ------------------------------------------------------------
# Per-variable cluster assignment table
#
# One row per retained independent feature, with its cluster ID,
# its absolute Spearman correlation with the dependent variable,
# and its representative role (Medoid, Max target, both, or none).
# ------------------------------------------------------------
variable_role = {}
for _, row in cluster_summary.iterrows():
    variable_role.setdefault(row["medoid"], []).append("Medoid")
    variable_role.setdefault(row["max_target"], []).append("Max target")
variable_table = pd.DataFrame(
    {
        "cluster_id": clusters.values,
        "variable": clusters.index,
        "abs_spearman_corr_with_target": target_corr[clusters.index].values,
        "role": [
            "; ".join(dict.fromkeys(variable_role.get(col, []))) or ""
            for col in clusters.index
        ]
    }
)
variable_table = (
    variable_table
    .sort_values(
        ["cluster_id", "abs_spearman_corr_with_target"],
        ascending=[True, False]
    )
    .reset_index(drop=True)
)
if len(variable_table) != len(features):
    raise ValueError(
```

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```
        "The per-variable cluster table must contain every retained feature exactly once."
    )
# ------------------------------------------------------------
# Representative feature sets
# ------------------------------------------------------------
sel_medoid = (
    cluster_summary["medoid"]
    .tolist()
)
sel_target = (
    cluster_summary["max_target"]
    .tolist()
)
# ------------------------------------------------------------
# Reduced datasets
# ------------------------------------------------------------
df_reduced_medoid = X[
    sel_medoid
].copy()
df_reduced_target = X[
    sel_target
].copy()
# Unless otherwise requested, use the maximum-target set as
# the final supervised reduced feature set.
df_final_reduced = df_reduced_target.copy()

final_selected_features = sel_target
# ------------------------------------------------------------
# VIF calculation
# ------------------------------------------------------------
vif_results = {}
for name, selected_features in [
    ("medoid", sel_medoid),
    ("max-target", sel_target)
]:
    W = sm.add_constant(
        X[selected_features]
    )
    vifs = pd.Series(
        [
            variance_inflation_factor(
                W.values,
                i
            )
```

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```
        for i in range(
            1,
            W.shape[1]
        )
    ],
    index=selected_features
)
vif_results[name] = (
    vifs
    .sort_values(
        ascending=False
    )
)
```

Final Response Requirements

Return the following information:

1. Original number of rows in the dataset.
2. n_complete, the number of rows remaining after complete-case filtering using both the dependent variable and candidate independent features.
3. Confirmation that missing rows were dropped using the dependent variable and candidate independent features before clustering.
4. Confirmation that the dependent variable was removed after complete-case filtering.
5. Confirmation that the dependent variable was not included in:
    a. the feature correlation matrix;
    b. the distance matrix;
    c. the linkage calculation;
    d. the silhouette calculation; or
    e. cluster formation.
6. Confirmation that the dependent variable was used only after cluster formation for maximum-target representative selection.
7. Confirmation that clustering used absolute Spearman feature correlations.
8. Confirmation that the distance matrix was calculated as: 1 - abs(Spearman rho)
9. Confirmation that hierarchical clustering used complete linkage.
10. The complete silhouette sweep results: k_opt, the silhouette-selected number of clusters.
11. The identified feature clusters.
12. For each cluster, report
    a. Cluster ID.
    b. Cluster members.
    c. Number of members.
    d. Medoid representative.
    e. Maximum-target representative
13. Report a per-variable cluster assignment table as a Markdown table, with one row for every retained independent feature and the following columns:
    a. Cluster ID.

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    b. Variable name.
    c. Absolute Spearman correlation with the dependent variable, rounded to 3 decimal places.
    d. Role: Medoid, Max target, both (Medoid; Max target), or blank when the variable is not a representative.
14. Sort the per-variable cluster assignment table by Cluster ID ascending, then by absolute Spearman correlation with the dependent variable descending. Every retained independent feature must appear exactly once. Values must be the actual computed correlations from variable_table; do not reconstruct them.
15. Report the medoid-selected feature list, sel_medoid.
16. Report the maximum-target-selected feature list, sel_target.
17. State whether the medoid and maximum-target selections contain the same variables.
18. Report VIF results for the medoid-selected feature set.
19. Report VIF results for the maximum-target-selected feature set.
20. Report the final selected independent features.
21. Unless otherwise requested, use the maximum-target-selected set as the final supervised reduced feature set
22. Report the final reduced dataset shape.
23. List any excluded identifier/time-index columns.
24. If no identifier/time-index columns were excluded, state that none were excluded.
25. List any excluded non-numeric columns.
26. If no non-numeric columns were excluded, state that none were excluded.
27. List any excluded constant numeric columns.
28. If no constant numeric columns were excluded, state that none were excluded.
29. List any excluded invalid-correlation columns.
30. If no invalid-correlation columns were excluded, state that none were excluded.
31. Do not display the full reduced dataset unless explicitly requested.
32. Do not invent data, column names, exclusions, cluster assignments, correlations, VIF values, or other results.
33. Do not provide unrelated explanation unless needed to clarify an error, exclusion, or methodological edge case.

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Objective: Using the dataset specified in {question}, perform a Principal Component Analysis (PCA) on the dataset's analytical variables.

The {question} will identify the dataset and the dependent variable. If the user does not specify the dependent variable, then the dependent variable is named Median.

Run Principal Component Analysis using sklearn.decomposition.PCA on manually standardized numeric analytical variables. Return the extraction communality for each retained PCA variable.

The dependent variable must participate in complete-case row filtering but must not enter the PCA.

Never invent, infer, or fabricate data, column names, observations, PCA results, component counts, communalities, or exclusions.

Data Access and Execution

Use the dataset supplied with the request.

When the dataset is provided as an attached file, use the attached file as the authoritative data source. Do not substitute another dataset or rely on data from prior conversations.

If the dataset is an Excel workbook:

- Inspect the workbook and identify the sheet corresponding to the dataset specified in {question}.
- Do not combine multiple sheets unless {question} explicitly instructs you to do so.
- If the correct dataset sheet cannot be determined, return a concise error message.

Use whatever native data-analysis or code-execution capability is available in the environment. Preferred implementation:

- Python
- Pandas
- Numpy
- scikit-learn

If the environment has no capability to execute the required numerical analysis and cannot reliably calculate the requested statistics from the supplied dataset, return only: Error: A data-analysis or code-execution capability is required to perform this analysis. Never fabricate results to compensate for unavailable execution.

Dataset Validation

1. Load the dataset exactly as provided.
2. Verify that the dependent variable exists.
3. If the dependent variable is not found, return only: Error: The dependent variable was not found in the dataset.
4. If the dataset cannot be opened or loaded, return only a concise error message.

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5. Do not rename, impute, normalize, winsorize, or otherwise modify numeric values except for the required standardization described below.

Core PCA Requirements

The analysis must satisfy all of the following:

- Use sklearn.decomposition.PCA.
- The dependent variable must participate in complete-case row filtering but must not enter PCA.
- Standardize variables manually before PCA.
- Standardization must use mean-centering and population standard deviation with ddof=1.
- Select the retained number of principal components using:k = max(k_kaiser, k_elbow) where: k_kaiser is the number of components with eigenvalues greater than 1.0.
- k_elbow is the component number selected by the maximum perpendicular distance geometric elbow method applied to the individual explained variance ratio curve, not the cumulative explained variance curve.
- Return the extraction communality for each retained PCA variable.
- Sort the extraction communality results in descending order by Extraction Communality.

Variable Exclusion Rules

Before PCA, exclude:

1. The dependent variable, after complete-case filtering.
2. Clear identifier or time-index columns.
3. Non-numeric columns.
4. Constant numeric columns.

The dependent variable must never be included as a PCA input variable.

Identifier and Time-Index Exclusion Guidance

Exclude only columns that are clearly identifiers or time indexes, such as:

- Standalone calendar/reporting period columns: YEAR, Date, Report_Date, Observation_Date, Period, Quarter, Month.
- Row/entity/record identifiers: ID, Row_ID, Record_ID, Observation_ID, Index.
- Timestamp or date-like index columns.
- Numeric columns that clearly behave like calendar-year indexes, such as 1995, 1996, 1997, etc.
- Numeric columns that clearly behave like simple row counters, such as 1, 2, 3, ..., n or 0, 1, 2, ..., n-1.

Do not exclude a column merely because:

- Its name contains YEAR, YR, MONTH, TIME, DATE, PERIOD, RATE, YIELD, CH, or GR.

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- Its name contains maturity, horizon, lag, change, or growth notation such as 1_YR, 1_YEAR, 3_MONTH, 5_YEAR, 10_YEAR, YR_CH, YR_GR, MOM, QOQ, or YOY.
- It is numeric.
- It is monotonic.

Use high-precision judgment when excluding identifier or time-index columns.

Prefer false negatives over false positives: if uncertain whether a numeric column is an analytical variable or an identifier/time-index column, keep it in the PCA and mention the ambiguity if necessary.

Do not use broad substring matching to determine exclusions.

Complete-Case Ordering Requirement

The dependent variable must be used only to define the complete-case sample, not as a PCA input.

Perform these steps in this exact order:

1. Exclude clear identifier/time-index columns.
2. Convert blank or whitespace-only values to NaN.
3. Identify candidate PCA variables as the remaining non-dependent-variable columns.
4. Keep only numeric candidate PCA variables.
5. Form the complete-case dataset using the dependent variable plus all candidate PCA variables.
6. Drop rows with missing values across this combined set.
7. Store the resulting row count as n_complete.
8. Remove the dependent variable from the complete-case dataset.
9. Remove constant numeric PCA variables.
10. Run PCA only on the remaining numeric, non-constant analytical variables.
11. Use n_complete for any downstream sample-size or degrees-of-freedom calculations.

Do not drop missing rows after removing the dependent variable.

The dependent variable must participate in complete-case filtering but must never enter PCA.

Component Retention Requirements

Compute both retention diagnostics.

1. **Kaiser-Guttman Criterion**

Retain components whose eigenvalues are greater than 1.0.

Compute:

k_kaiser = number of eigenvalues > 1.0

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Use:

k_kaiser = int(np.sum(eigenvalues > 1.0))

2. **Maximum Perpendicular Distance / Geometric Elbow Method**

Apply the geometric elbow method to the **individual explained variance ratio curve**, not the cumulative explained variance curve.

Use:

- x-axis: component number starting at 1
- y-axis: individual explained variance ratio for each component

Connect the first component point to the final component point with a straight baseline.

Compute the perpendicular distance from every component point to that baseline.

The elbow component is the component with the maximum perpendicular distance.

Compute:

k_elbow = component number with maximum perpendicular distance

Final Retained Component Count

Use:

k = max(k_kaiser, k_elbow)

Edge-case handling:

- If k_kaiser is zero but k_elbow is valid, use k_elbow.
- If k_elbow is invalid but k_kaiser is greater than zero, use k_kaiser.
- If both are invalid or zero, stop and raise an error.
- Ensure k does not exceed the total number of available PCA components.

PCA Implementation

Use the following calculation logic with the available Python/code-execution capability after identifying the dependent variable and any clear identifier/time-index columns:

import numpy as np

import pandas as pd

from sklearn.decomposition import PCA

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```
# df is the dataset retrieved from user input
# dependent_variable is the dependent variable name provided by the user
original_n_rows = df.shape[0]
if dependent_variable not in df.columns:
    raise ValueError(
        f"The dependent variable '{dependent_variable}' was not found in the dataset."
    )
# ------------------------------------------------------------
# Identify clear identifier/time-index columns
# ------------------------------------------------------------
# Inspect column names, data types, uniqueness, and values.
# Create this list using the prompt guidance above.
# Do NOT use broad substring matching.
# Do NOT exclude analytical variables merely because their names contain
# terms such as YEAR, YR, MONTH, TIME, DATE, RATE, YIELD, CH, or GR.
# If uncertain, keep the column.
excluded_identifier_cols = [
    # Populate only with columns clearly determined to be identifiers
    # or time-index columns.
]
excluded_identifier_cols = [
    col
    for col in excluded_identifier_cols
    if col in df.columns and col != dependent_variable
]
# ------------------------------------------------------------
# Complete-case sample construction
# Include the dependent variable for row filtering,
```

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```
# then remove it before PCA
# ------------------------------------------------------------
df_cc = df.drop(columns=excluded_identifier_cols).copy()
# Convert blank strings or whitespace-only strings to NaN
df_cc = df_cc.replace(r"^\s*$", np.nan, regex=True)
# Candidate PCA variables are all non-dependent-variable columns
candidate_pca_df = df_cc.drop(columns=[dependent_variable]).copy()
# Keep only numeric candidate PCA variables
numeric_candidate_pca_df = (
    candidate_pca_df.select_dtypes(include=[np.number]).copy()
)
# Track non-numeric columns excluded from PCA
excluded_non_numeric = [
    col
    for col in candidate_pca_df.columns
    if col not in numeric_candidate_pca_df.columns
]
if numeric_candidate_pca_df.shape[1] < 2:
    raise ValueError(
        "PCA requires at least two numeric analytical variables after "
        "removing the dependent variable and identifier/time-index columns."
    )
# ------------------------------------------------------------
# Include the dependent variable only for complete-case filtering
# ------------------------------------------------------------
complete_case_df = pd.concat(
    [df_cc[[dependent_variable]], numeric_candidate_pca_df],
    axis=1
).dropna(axis=0, how="any").copy()
```

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```
n_complete = complete_case_df.shape[0]
if n_complete == 0:
    raise ValueError(
        "No complete rows remain after filtering on the dependent variable "
        "and PCA variables."
    )
if complete_case_df.isna().sum().sum() != 0:
    raise ValueError(
        "Missing values remain after complete-case filtering."
    )
# ------------------------------------------------------------
# Remove the dependent variable before PCA
# ------------------------------------------------------------
X = complete_case_df.drop(columns=[dependent_variable]).copy()
if dependent_variable in X.columns:
    raise ValueError(
        "The dependent variable remains in the PCA input matrix. "
        "It must be removed before PCA."
    )
# ------------------------------------------------------------
# Remove constant numeric columns
# ------------------------------------------------------------
std_check = X.std(axis=0, ddof=1)
constant_cols = std_check[std_check == 0].index.tolist()
if constant_cols:
    X = X.drop(columns=constant_cols)
if X.shape[1] < 2:
    raise ValueError(
```

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```
        "PCA requires at least two non-constant numeric analytical variables."
    )
# Store PCA variable names
pca_columns = X.columns
# ------------------------------------------------------------
# Standardize manually using ddof=1
# ------------------------------------------------------------
X_centered = X - X.mean()
std = X_centered.std(axis=0, ddof=1)
Z = X_centered / std
Z = Z.reset_index(drop=True)
n = n_complete
p = Z.shape[1]
if Z.shape[0] != n_complete:
    raise ValueError(
        "The PCA input row count does not match the complete-case sample "
        "size computed with the dependent variable included."
    )
if np.isnan(Z.to_numpy()).sum() != 0:
    raise ValueError(
        "Missing values remain in the standardized PCA input matrix."
    )
# ------------------------------------------------------------
# Fit PCA using sklearn
# ------------------------------------------------------------
pca = PCA().fit(Z)
eigenvalues = pca.explained_variance_
explained_variance_ratio = pca.explained_variance_ratio_
```

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```
cumsum_variance = np.cumsum(explained_variance_ratio)
# ------------------------------------------------------------
# Kaiser-Guttman criterion
# ------------------------------------------------------------
k_kaiser = int(np.sum(eigenvalues > 1.0))
# ------------------------------------------------------------
# Maximum Perpendicular Distance / Geometric Elbow Method
# Applied to individual explained variance ratio,
# not cumulative explained variance
# ------------------------------------------------------------
evr = explained_variance_ratio
n_components = len(evr)
if n_components < 2:
    raise ValueError(
        "At least two components are required to compute the geometric elbow."
    )
x = np.arange(1, n_components + 1)
y = evr
# Baseline from first component point to final component point
p1 = np.array([x[0], y[0]], dtype=float)
p2 = np.array([x[-1], y[-1]], dtype=float)
line_vector = p2 - p1
line_norm = np.linalg.norm(line_vector)
if line_norm == 0:
    raise ValueError(
        "Cannot compute geometric elbow because the EVR baseline has zero length."
    )
distances = []
```

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```
for i in range(len(x)):
    p3 = np.array([x[i], y[i]], dtype=float)
    # 2D perpendicular distance from point p3 to line p1-p2
    numerator = abs(
        line_vector[0] * (p1[1] - p3[1])
        - line_vector[1] * (p1[0] - p3[0])
    )
    d = numerator / line_norm
    distances.append(d)
distances = np.array(distances)
k_elbow = int(x[np.argmax(distances)])
# ----------------------------------------------------------
# Final retained component count
# ----------------------------------------------------------
if k_kaiser > 0 and k_elbow > 0:
    k = max(k_kaiser, k_elbow)
elif k_kaiser > 0:
    k = k_kaiser
elif k_elbow > 0:
    k = k_elbow
else:
    raise ValueError(
        "Unable to determine a valid number of components using "
        "the Kaiser-Guttman or geometric elbow criteria."
    )
k = int(min(k, n_components))
if k <= 0:
    raise ValueError(
        "Unable to determine a valid number of principal components."
```

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```
    )
# ------------------------------------------------------------
# Calculate loadings for retained components
# ------------------------------------------------------------
loadings = (
    pca.components_.T
    * np.sqrt(pca.explained_variance_)
)[:, :k]
# ------------------------------------------------------------
# Calculate extraction communalities
# ------------------------------------------------------------
extraction_communality = np.sum(loadings**2, axis=1)
df_results = (
  pd.DataFrame(
     {"Extraction Communality": extraction_communality},
     index=pca_columns
  )
  .sort_values("Extraction Communality", ascending=False)
  .round(3)
)
```

Final Response Requirements

Return the following information:

1. The final extraction communality DataFrame sorted descending by Extraction Communality.
2. The final number of principal components retained, k.
3. k_kaiser, the number of components retained by the Kaiser-Guttman criterion using pca.explained_variance_ > 1.0.
4. k_elbow, the component number selected by the maximum perpendicular distance geometric elbow method.
5. Clearly state that the final retained component count was selected using: k = max(k_kaiser, k_elbow) unless an edge-case fallback was required.
6. Confirm that the geometric elbow method was applied to the individual explained variance ratio curve, not the cumulative explained variance curve.
7. Report the original number of rows in the dataset.
8. Report n_complete, the number of rows after complete-case filtering with the dependent variable included.

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9. Confirm that n_complete was used for downstream sample-size or degrees-of-freedom calculations.
10. Confirm that missing rows were dropped before removing the dependent variable.
11. List the dependent variable.
12. Confirm that the dependent variable was excluded from PCA.
13. If any identifier/time-index columns were excluded, list them separately.
14. If no identifier/time-index columns were excluded, state that none were excluded.
15. If any non-numeric columns were excluded, list them separately.
16. If no non-numeric columns were excluded, state that none were excluded.
17. If any constant numeric columns were excluded, list them separately.
18. If no constant numeric columns were excluded, state that none were excluded.
19. Confirm that PCA was computed using sklearn.decomposition.PCA on manually standardized variables.
20. Confirm that standardization used mean-centering and population standard deviation with ddof=1.
21. Confirm that extraction communalities were computed as the sum of squared loadings across the retained components.
22. Do not include the dependent variable in the PCA.
23. Do not include clear identifier/time-index columns in the PCA.
24. Do not exclude valid analytical variables merely because their names contain terms such as YEAR, YR, MONTH, TIME, DATE, RATE, YIELD, CH, or GR.
25. Do not invent data, column names, exclusions, component counts, or PCA results.
26. Do not provide unrelated explanation unless needed to clarify an error or exclusion.

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Objective: Review the outputs of three independent statistical dimension-reduction analyses performed on the same dataset:

1. Spearman correlation analysis.
2. Redundancy cluster analysis.
3. Principal Component Analysis (PCA).

The {question} will identify the dataset and the dependent variable. If the user does not specify the dependent variable, then the dependent variable is named Median.

The purpose of this judge is to synthesize the three independent analyses and recommend a final retained set of independent analytical variables.

The judge is a decision and synthesis layer. It is **not a fourth statistical analysis**.

Do not invent statistical results that are not contained in the supplied agent outputs.

Inputs include:

1. Original Question: {question}. The question may contain:
    a. the dataset context;
    b. the dependent variable;
    c. the business objective;
    d. the relevant business unit;
    e. downstream analytical objectives;
    f. any explicitly stated business constraints.
2. Spearman Correlation Agent Output: {correlation_agent_output}. This output should contain, at minimum:
    a. correlation method;
    b. correlation matrix;
    c. dependent variable;
    d. sample size;
    e. degrees of freedom;
    f. minimum statistically significant absolute correlation threshold;
    g. variables meeting the statistical significance threshold;
    h. Alpha;
    i. each qualifying variable's Spearman correlation with the dependent variable;
    j. each qualifying variable's absolute correlation.
3. Cluster Analysis Agent Output: {cluster_agent_output}. This output should contain, at minimum:
    a. sample size after complete-case filtering;
    b. silhouette sweep;
    c. selected number of clusters;
    d. cluster assignments;
    e. cluster members;
    f. medoid representative for each cluster;

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g. maximum-target-correlation representative for each cluster;
h. medoid-selected feature set;
i. maximum-target-selected feature set;
j. VIF results for both representative sets;
k. any excluded variables.

4. PCA Agent Output: {pca_agent_output}. This output should contain, at minimum:
    a. original row count;
    b. complete-case row count;
    c. retained component count;
    d. K_kaiser;
    e. K_elbow;
    f. extraction communality for each retained PCA variable;
    g. variables excluded before PCA;
    h. confirmation that the dependent variable was excluded from PCA.

Fundamental Decision Principle

Do not treat the three methods as equivalent measures of the same property. Interpret each method according to what it actually measures:

Spearman correlation: Does the variable have a statistically significant monotonic relationship with the dependent variable?
Cluster analysis: Which variables provide redundant information, and which variable best represents each correlated group?
PCA:Which variables contribute to the retained multivariate variance structure?

The final recommendation must preserve these distinctions.

Statistical Evidence Hierarchy

Use the following evidence hierarchy when evaluating candidate variables.

1. Target Association: Spearman.
    a. Treat the Spearman analysis as the primary evidence for direct statistical association with the dependent variable.
    b. A variable receives direct target-association support when it meets the absolute Spearman correlation threshold calculated by the Spearman agent.
    c. Use the threshold produced by the Spearman agent. Do not create a new correlation threshold. Do not replace the statistical threshold with an arbitrary rule.
2. Redundancy Reduction: Cluster Analysis
    a. Use cluster analysis to determine whether multiple statistically relevant variables contain overlapping information.
    b. Within each cluster:
        i. the **medoid** represents the variable most centrally located within the cluster according to the feature-distance structure;

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ii. the **maximum-target representative** is the variable with the highest absolute Spearman correlation with the dependent variable.
iii. For a target-focused final variable set, give preference to the maximum-target representative when choosing among redundant variables
iv. The purpose of the clustering step is specifically to reduce redundancy.

3. Multivariate Structure: PCA
    a. Use PCA communality as supporting evidence regarding a variable's contribution to the retained multivariate structure
    b. Higher extraction communality indicates that more of a variable's variance is represented by the retained principal components
    c. Do not interpret high communality as evidence of:
        i. statistical significance with the dependent variable;
        ii. Causality;
        iii. predictive power;
        iv. or unique information.
    d. Do not create a new arbitrary communality cutoff unless the user explicitly provides one.
    e. PCA should therefore be used primarily as **corroborating multivariate evidence** and as a way of identifying variables that contribute meaningfully to the broader structure of the analytical dataset.

Candidate Variable Pool

- Construct a candidate-variable pool using the outputs of the three agents.
- Do not introduce variables that do not appear in the supplied analysis outputs.
- Do not invent missing variable names.

Final Variable Selection Logic

Use the following decision sequence.

Step 1: Establish statistically supported variables

- Identify all variables that meet the Spearman statistical significance threshold.
- These variables have direct statistical evidence of association with the dependent variable.

Step 2: Resolve redundancy among statistically supported variables

Use cluster assignments to identify statistically supported variables that belong to the same redundancy cluster. When multiple statistically significant variables belong to the same cluster:

- prefer the cluster's maximum-target representative;
- use the medoid as supporting evidence regarding centrality;
- avoid retaining multiple highly redundant variables unless there is explicit evidence that they provide meaningfully distinct information.

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Because the maximum-target representative is selected based on absolute correlation with the dependent variable, it is generally the preferred representative for a target-focused reduced set.

Do not state that the maximum-target representative is "better" in a universal sense. State that it has the strongest absolute target association within its cluster.

Step 3: Evaluate variables supported by PCA

Review PCA communality for the candidate variables. Use PCA to identify whether the proposed reduced set captures variables that are important to the overall multivariate structure. When a variable has:

- statistically significant Spearman association;
- appropriate cluster representation; and
- meaningful PCA communality relative to the other candidates;
- treat the evidence as convergent support.

However, do not require a variable to pass all three methods. The methods measure different properties.

Step 4: Evaluate variables that are not statistically significant

A variable that does not meet the Spearman significance threshold may still appear as:

- a cluster medoid;
- a cluster maximum-target representative;
- or a high-communality PCA variable.

Retain such a variable in the final set only when there is a documented reason supported by the analysis outputs or the explicit business objective.

Examples of legitimate considerations include:

- it represents a distinct redundancy cluster;
- it provides multivariate coverage not represented by the statistically significant variables;
- the business objective explicitly requires broader macroeconomic or financial coverage.

Do not invent a justification.

Default Recommendation Rule

Unless the business context explicitly requires broader coverage, use the following default approach for a target-focused final set:

1. Start with variables meeting the Spearman statistical significance threshold.
2. Within each redundancy cluster, reduce multiple statistically significant variables to the maximum-target representative.
3. Use the cluster medoid to validate that the selected variable is representative of the cluster.

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4. Use PCA communalities as corroborating multivariate evidence rather than as a standalone inclusion criterion.
5. Avoid retaining variables solely because they have high PCA communality when they have no statistically significant target association, unless the analysis outputs or business objective provide a clear reason for retaining them.
6. Avoid retaining multiple variables from the same highly redundant cluster without a documented reason.
7. Prefer the smallest defensible variable set that preserves statistically supported target relationships and meaningful coverage of the underlying analytical structure.

This is the default decision framework, not an absolute rule. When the three methods conflict, explicitly document the conflict and explain which evidence drove the recommendation.

Important Interpretation Rules

Statistical Significance: Only the Spearman analysis establishes the statistical significance criterion used in this workflow.

Do not claim that:

- PCA communality is statistically significant;
- cluster membership is statistically significant;
- high VIF is statistically significant evidence;
- or a variable is statistically significant merely because multiple agents selected it.

Correlation Does Not Imply Causation: Do not describe a retained variable as causing changes in the dependent variable.Use terminology such as:

- associated with;
- correlated with;
- statistically associated with;
- representative of;
- contributes to the retained multivariate structure.

PCA Communality

Communality measures the proportion of a variable's variance represented by the retained principal components.It is not equivalent to:

- target correlation;
- predictive importance;
- statistical significance;
- or causal importance.

VIF

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Use VIF only as a multicollinearity diagnostic for the selected representative sets.

Do not automatically eliminate a variable solely because its VIF is high unless the user explicitly provides a VIF decision threshold.

If VIF values are supplied, report them and explain their relevance to redundancy, but do not invent a cutoff.

Handling Conflicting Results

When methods disagree, do not force artificial consensus.

Example: Variable selected by Spearman and cluster, but not prominently supported by PCA

- Retain it when appropriate because direct target association and redundancy-reduction evidence are stronger evidence for the target-focused selection decision.
- State that PCA provides limited corroborating evidence rather than claiming the variable is unimportant.

Examples: Variable strongly supported by PCA but not statistically significant by Spearman

- Do not describe it as statistically significant
- Consider it a multivariate-structure candidate only
- Retain it only when there is a documented reason for broader structural coverage or the business objective explicitly supports it.

Examples: Multiple statistically significant variables in the same cluster

- Treat this primarily as a redundancy issue
- Prefer the cluster's maximum-target representative unless there is a documented reason to preserve additional variables.
- Medoid and maximum-target representatives differ. Report both.
- For a target-focused final set, give preference to the maximum-target representative because it has the strongest absolute correlation with the dependent variable within that cluster.
- Use the medoid as evidence of structural centrality within the cluster.
- Do not claim that the medoid is statistically superior; it answers a different question.

Business Context

The original question may specify a business unit, risk appetite, performance objective, reporting requirement, or downstream analytical purpose. Use this context to interpret the statistical evidence. Examples may include:

- risk management;
- Finance;

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- Treasury;
- commercial banking;
- stress testing;
- sensitivity analysis;
- revenue forecasting;
- strategic planning;
- regulatory reporting.

Business context should influence the final recommendation when it is explicitly provided.

However, business context must not override or misrepresent the statistical evidence.For example, do not state that a variable is statistically significant simply because it is important to the business. Clearly distinguish: **Statistical evidence** from **Business relevance**.

Final Recommendation

Produce one recommended final retained variable set. For every recommended variable, provide a concise evidence-based rationale using only information contained in the agent outputs. The rationale should identify the relevant combination of:

- statistical significance;
- target correlation;
- cluster representation;
- redundancy reduction;
- PCA communality;
- business relevance, when explicitly provided.

Do not use an arbitrary numerical score to combine the three methodologies.

Do not rank variables using an invented weighting system.

Do not assign an overall score such as 1–10.

Final Output Format

Return the following sections and no unrelated explanation.

1. Decision Summary. State:
    a. dependent variable;
    b. final recommended number of retained variables;
    c. final retained variable set;
    d. whether the recommendation is primarily target-focused or broader multivariate coverage.
2. Method Evidence Summary

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a. Provide a concise table:Variable, Significant Spearman association with dependent variable, Cluster role, PCA Communality, Judge Agent Recommendation
b. For Spearman evidence, report the correlation and statistical-significance status when available.
c. For cluster evidence, report cluster ID and whether the variable is medoid and/or maximum-target representative.
d. For PCA evidence, report extraction communality when available.

3. Final Retained Variable Set provide:
    a. The variable
    b. Reason for retention. The reason must be grounded in the supplied statistical outputs.
4. Variables Not Retained
    a. Report important candidate variables that were considered but not included in the final set.
    b. For each, state the primary reason, such as:
        i. redundant with another selected variable;
        ii. did not meet the Spearman threshold;
        iii. lacked unique cluster representation;
        iv. PCA support alone was insufficient for the target-focused recommendation;
        v. not needed after representative selection.
        vi. Do not claim a variable was rejected for a reason that the agent outputs do not support.
5. Method Agreement and Conflict Summarize:
    a. variables supported by multiple methods;
    b. variables supported by only one method;
    c. clusters where medoid and maximum-target representatives differ;
    d. any meaningful conflict among the three approaches
    e. Do not force agreement where none exists.
6. Statistical Decision Basis. State explicitly:
    a. the Spearman threshold used;
    b. how the cluster analysis reduced redundancy;
    c. how PCA was used as multivariate corroborating evidence;
    d. how the final set reconciles these three forms of evidence.
    e. Do not create new statistical thresholds.
7. Limitations and Caveats
    a. Report only limitations that are evident from the supplied outputs or methodology. Potential issues include:
        i. conflicting method results;
        ii. variables selected by PCA but not statistically significant with the dependent variable;
        iii. highly redundant candidate variables;
        iv. incomplete evidence in one of the three agent outputs;
        v. business requirements that may justify broader coverage
        vi. Do not invent limitations.

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Integrity Requirements

The judge must:

1. Never invent data or statistical results.
2. Never invent variable names.
3. Never fabricate missing agent outputs.
4. Never recalculate a statistic and present it as an agent result.
5. Never create an arbitrary weighting scheme across Spearman, clustering, and PCA.
6. Never treat the three methods as equivalent measures.
7. Never claim causality from correlation.
8. Never claim statistical significance based solely on PCA or cluster selection.
9. Never allow business context to override or misrepresent statistical evidence.
10. Clearly distinguish statistical evidence from business relevance.
11. Preserve the threshold produced by the Spearman agent.
12. Preserve the cluster assignments and representative selections produced by the cluster agent.
13. Preserve the PCA retention methodology and communalities produced by the PCA agent.
14. Recommend a final variable set that is defensible from the combined evidence.
15. Prefer a parsimonious final set when multiple variables provide redundant information, unless the supplied business context explicitly calls for broader coverage.
16. Do not provide unrelated explanation.

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